

Site-1

Site-1 is your **reference branch**—a small but complete environment containing:

- **1 Router** (acts as the site's edge device, gateway to WAN)
- **1 Switch** (distribution + access for local VLANs)
- **2 VLANs**
 - **Users VLAN** → for workstations
 - **Servers VLAN** → for local services (file, DHCP, maybe later a local DNS)

◇ Day-1: Addressing & Documentation

Your task says:

1. Add an addressing table for Site-1 in your documentation.
2. Define the base network (e.g., 192.168.10.0/24).

Step-by-Step Breakdown

1. Define the Base Network Block

Pick a **base private network** for this site.

For example, 192.168.10.0/24 supports up to 254 usable host IPs.

Why:

Each site should have its own unique subnet block to avoid overlapping when you interconnect sites later (important for routing).

2. Plan VLAN Segments

Split the /24 into smaller, logically separated subnets per VLAN.

VLAN	Purpose	Subnet	Gateway (default) (default gateway = first usable IP of each subnet).	Notes
VLAN 10	Users	192.168.10.0/26	192.168.10.1	62 usable IPs
VLAN 20	Servers	192.168.10.64/26	192.168.10.65	62 usable IPs

You can reserve:

- **.1 and .65** as the *router's SVI (default gateway)* for each VLAN.
- **.2–.50** for hosts in User VLAN.
- **.66–.100** for servers.

Why:

Subnetting isolates broadcast domains, improves security and traffic management, and prepares you for routing between VLANs.

3. Allocate Router and Switch Interfaces

- The router will have:
 - One *LAN interface* for the switch trunk (802.1Q sub-interfaces for VLAN routing).
 - One *WAN interface* (future use for HQ link).
- The switch will have:
 - VLAN 10 and VLAN 20 are configured.
 - Trunk port toward the router.
 - Access ports for user PCs and servers.

Why:

This models a *Router-on-a-Stick* topology — the simplest way to perform inter-VLAN routing on a small site.

4. Assign IPs to End Hosts

Decide which IPs to go to PCs and servers.

Device	VLAN	Interface	IP	Gateway
PC1	VLAN 10 (Users)	Fa0/1	192.168.10.10	192.168.10.1
PC2	VLAN 10 (Users)	Fa0/2	192.168.10.11	192.168.10.1
Server1	VLAN 20 (Servers)	Fa0/3	192.168.10.70	192.168.10.65

5. Addressing Table

Example Table for Site-1

Device	Interface	VLAN	IP Address	Subnet Mask	Gateway	Description
Router R1	G0/0.10	VLAN 10	192.168.10.1	255.255.255.192	–	Users VLAN gateway
Router R1	G0/0.20	VLAN 20	192.168.10.65	255.255.255.192	–	Servers VLAN gateway
Switch S1	VLAN 1 (mgmt)	–	192.168.10.254	255.255.255.192	192.168.10.1	Management
PC1	Fa0/1	VLAN 10	192.168.10.10	255.255.255.192	192.168.10.1	User workstation
Server1	Fa0/3	VLAN 20	192.168.10.70	255.255.255.192	192.168.10.65	File server

VLAN and Port Configuration on the Switch

You're essentially telling the switch, "These ports belong to VLAN 10, these to VLAN 20, and the router link should carry both VLANs as a trunk."

S1 Configuration (Cisco-style)

```
enable
configure terminal
!
vlan 10
  name USERS
vlan 20
```

```

name SERVERS
!
interface fastEthernet0/1
  switchport mode access
  switchport access vlan 10
  description User_PC1
!
interface fastEthernet0/2
  switchport mode access
  switchport access vlan 20
  description Server_1
!
interface gigabitEthernet0/1
  switchport mode trunk // or we can use allowed vlans
  description Link_to_Router_R1
!
exit

```

Explanation:

- Access ports are end devices, tagged internally.
- Trunk = carries multiple VLANs toward the router using *802.1Q encapsulation*.
- VLAN naming is just documentation of sugar, but it is critical for clarity later.

◇ Step 3: Router-on-a-Stick Configuration

The router has a single physical interface (say, `g0/0`) connected to S1's trunk port. You'll carve that interface into **sub-interfaces**, one per VLAN.

R1 Configuration

```

enable
configure terminal
!
interface gigabitEthernet0/0
  no shutdown
!

```

```
interface gigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.192
  description Users_VLAN
!
interface gigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.10.65 255.255.255.192
  description Servers_VLAN! exit
```

Explanation:

- Encapsulation dot1Q tags the VLAN traffic.
- A router routes between VLANs internally—that's inter-VLAN routing.
- Each sub-interface acts as the **default gateway** for its VLAN.

◇ Step 4: Configure Hosts

Set static IPs for now:

- **PC1**
 - IP: 192.168.10.10
 - Mask: 255.255.255.192
 - Gateway: 192.168.10.1
- **Server1**
 - IP: 192.168.10.70
 - Mask: 255.255.255.192
 - Gateway: 192.168.10.65

◇ Step 5: Verification —"Progress Criteria"

Step 5.1 – Basic checks

On **R1**:

```
show ip interface brief
show vlan
```

Confirm both sub-interfaces are *up/up*, and VLANs exist.

Step 5.2 – Ping tests

From **PC1**, ping:

- Its own gateway → ping 192.168.10.1
- Other VLAN gateway → ping 192.168.10.65
- Server → ping 192.168.10.70

If all succeed—congratulations! That's **an inter-VLAN routing functionality**.

If the cross-VLAN ping fails:

- Check that the switch trunk port is correctly set (show interfaces trunk).
- Verify that both sub-interfaces are up.
- Confirm no IP overlap or wrong mask.

WEEK-2 DOCUMENTATION —IPv6

Addressing & Dual Stack Setup

1. Overview

This week focuses on implementing IPv6 addressing within the Site-1 network. A dual-stack environment is introduced by adding IPv6 addresses alongside the existing IPv4 plan. Tasks include defining the IPv6 addressing scheme, configuring IPv6 on network devices, validating connectivity, and extending Python automation to support IPv6.

2. Objectives

- Define a structured IPv6 addressing plan for Site-1.
- Configure IPv6 on the router, switches, and end devices.
- Verify end-to-end IPv6 connectivity.
- Update the Week-1 Python script to support IPv6 network calculations.

3. IPv6 Addressing Plan

3.1 Assigned IPv6 Prefixes

VLAN	Purpose	Prefix	Gateway Address
10	User Devices	2001:DB8:10::/64	2001:DB8:10::1
20	Server Network	2001:DB8:20::/64	2001:DB8:20::1
WAN IPv6 subnet		2001:db8:100::/64	

4. Device Configurations

4.1 Router Configuration (R1)

```
ipv6 unicast-routing
```

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ipv6 address 2001:DB8:10::1/64
```

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ipv6 address 2001:DB8:20::1/64
```

4.2 Switch Configuration (SW1)

(Access ports previously assigned in Week 1; no IPv6-specific changes required.)

Example confirmation:

```
interface Fa0/1
  switchport mode access
  switchport access vlan interface Fa0/2
  switchport mode access
  switchport access vlan interface Fa0/3
  switchport mode access
  switchport access vlan 20
```

4.3 Host & Server Configurations

PC1 (VLAN10)

- IPv6 Address: 2001:DB8:10::10
- Prefix: /64
- Gateway: 2001:DB8:10::1

Laptop1 (VLAN10)

- IPv6 Address: 2001:DB8:10::11
- Prefix: /64
- Gateway: 2001:DB8:10::1

Server1 (VLAN20)

- IPv6 Address: 2001:DB8:20::20
- Prefix: /64
- Gateway: 2001:DB8:20::1

5. Verification Steps

5.1 Router & Host Reachability

From hosts:

```
ping 2001:DB8:10::1
```

From server:

```
ping 2001:DB8:20::1
```

5.2 Inter-VLAN IPv6 Connectivity

```
ping 2001:DB8:20::20      (from VLAN10)
ping 2001:DB8:10::10      (from VLAN20)
```

5.3 Neighbor Discovery Check

```
show ipv6 neighbors
show ipv6 interface brief
```

6. Python Automation Update

The IPv6-enhanced script performs:

- IPv6 network parsing
- Gateway suggestion generation
- Host range representation
- Integrated IPv4 + IPv6 output formatting

Include the updated script in the appendix or “Artifacts” section.

7. Results & Observations

- IPv6 successfully enabled on all network segments.
- All devices reachable via IPv6 within their VLANs.
- Dual-stack configuration functions correctly.
- Python tool updated to support IPv6 calculations.

8. Issues Encountered (If Any)

Document problems such as:

- Incorrect prefix assignment
- Host OS not enabling IPv6
- Subinterface configuration errors
- VLAN misconfiguration
- on

Include how each issue was resolved.

9. Conclusion

IPv6 addressing has been fully deployed across Site-1, establishing a dual-stack environment. This lays the groundwork for upcoming tasks involving inter-VLAN routing, DHCPv6, SLAAC, and advanced automation



WEEK-3 STP and PORT SECURITY

STP Overview (Site-1)

- **STP Mode:** PVST+
- **Root Bridge:** SW1 (forced via priority configuration)

- **VLANs Covered:** 10 (USERS), 20 (SERVERS), 999 (NATIVE)

Root Bridge Configuration Rationale

SW1 was intentionally configured as the root bridge for all VLANs to ensure deterministic topology behavior and avoid accidental root elections in future expansions.

STP State Summary

VLAN	Root Bridge	Forwarding Ports	Blocked Ports
10	SW1	Fa0/1, Fa0/3, Gi0/1	None
20	SW1	Fa0/2, Gi0/1	None
999	SW1	Gi0/1	None

No topological changes were detected. STP is stable.

STP Hardening Measures Applied

- PortFast enabled globally on access ports
- BPDU Guard enabled globally to protect against rogue switches
- Explicit root bridge priority configuration
- Default STP timers retained (Hello 2s, MaxAge 20s, ForwardDelay 15s)

Planned Enhancements

- Loop Guard to be enabled in multi-switch topology scenarios (Later because Packet tracer doesn't support it)

Port Security Policy

Interface	VLAN	Max MAC	Mode	Purpose
Fa0/1	10	1	Restrict	User access with monitoring
Fa0/2	20	1	Shutdown	High-security endpoint
Fa0/3	10	1	Shutdown	Strict access port

Sticky MAC Learning

Sticky MAC addresses are dynamically learned and saved in the running configuration to prevent unauthorized device access.

Scenario 1: Port Security Violation (Shutdown Mode)

- Trigger: Unknown MAC address connected
- Result: Port enters err-disabled state
- Recovery:

```
interface fa0/2
shutdown
no shutdown
```

Scenario 2: STP Priority Change

- Modified bridge priority
- Observed root recalculation
- Restored intended priority to reassert control

What I Implemented

- Forced STP root bridge
- Enabled PortFast and BPDU Guard
- Secured access ports using sticky MAC
- Differentiated security levels using restrict vs shutdown modes

What I Tested

- STP stability
- Port security violations
- Recovery procedures

What I Learned

- STP behavior is predictable when priorities are controlled
 - Port security must balance availability and protection
 - Documentation is as important as configuration
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✚ Site-2

Site-2 Topology

Logical Goal of Site-2

- Represent a **branch office**
- Same user/server separation
- Connected to Site-1 via WAN
- Designed so routing decisions actually matter

Site-2 – Physical Topology

Devices

- 1 × Router (R2 – 1941)
- 1 × Switch (SW2 – 2960)
- 2 × End devices

Site-2 Addressing and Routing Documentation

1. Site-2 Overview

Site-2 is a branch site connected to Site-1 via a routed WAN link.

The site uses VLAN-based segmentation with dual-stack (IPv4 + IPv6) addressing.

2. VLAN and Subnet Summary

VLAN Mapping

VLAN ID	Name	IPv4 Subnet	IPv6 Prefix
30	USERS_SITE2	192.168.20.0/26	2001:db8:30::/64
40	SERVER_SITE2	192.168.20.64/26	2001:db8:40::/64

3. Site-2 Addressing Table

Router R2 Interfaces

Device	Interface	VLAN	IPv4 Address	IPv6 Address	Description
R2	G0/0.30	30	192.168.20.1/26	2001:db8:30::1/64	Users gateway
R2	G0/0.40	40	192.168.20.65/26	2001:db8:40::1/64	Servers gateway
R2	G0/1	WAN	10.0.0.2/30	2001:db8:100::2/64	Link to Site-1

4. WAN Link Summary

Link	IPv4 Subnet	IPv6 Subnet
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R1 ↔ R2	10.0.0.0/30	2001:db8:100::/64
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5. Static Routing Configuration (R2)

IPv4 Static Routes

Destination Network	Next Hop	Purpose
192.168.10.0/26	10.0.0.1	Site-1 Users
192.168.10.64/26	10.0.0.1	Site-1 Servers

IPv6 Static Routes

Destination Prefix	Next Hop	Purpose
2001:db8:10::/64	2001:db8:100::1	Site-1 Users
2001:db8:20::/64	2001:db8:100::1	Site-1 Servers

6. Routing Decision Table (R2)

This table represents how Router R2 forwards traffic based on destination.

IPv4 Routing Decisions

Destination Network	Route Type	Next Hop	Outgoing Interface
192.168.20.0/26	Connected	—	G0/0.30
192.168.20.64/26	Connected	—	G0/0.40
192.168.10.0/26	Static	10.0.0.1	G0/1

192.168.10.64/26	Static	10.0.0.1	G0/1
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IPv6 Routing Decisions

Destination Prefix	Route Type	Next Hop	Outgoing Interface
2001:db8:30::/64	Connected	—	G0/0.30
2001:db8:40::/64	Connected	—	G0/0.40
2001:db8:10::/64	Static	2001:db8:100::1	G0/1
2001:db8:20::/64	Static	2001:db8:100::1	G0/1

7. Verification Commands

```
show ip route
show ipv6 route
ping <remote IPv4 address>
ping ipv6 <remote IPv6 address>
```

Router-on-a-Stick – Site-2 (IPv6)

On R2:

```
ipv6 unicast-routing
```

```
interface g0/0
no shutdown
```

```
interface g0/0.30
encapsulation dot1Q 30
ipv6 address 2001:DB8:30::1/64
```

```
interface g0/0.40
encapsulation dot1Q 40
```


ipv6 address 2001:DB8:40::1/64

(Optional later: enable IPv6 OSPFv3 here — not yet.)

End Devices – Examples

PC2 (VLAN 30)

IPv6 address: 2001:DB8:30::10/64

Gateway: 2001:DB8:30::1

Server2 (VLAN 40)

IPv6 address: 2001:DB8:40::10/64

Gateway: 2001:DB8:40::1

Interview-Grade Explanation (Memorize This)

“Each site has its own IPv6 prefix space to avoid overlap. I aligned VLAN IDs with the third hexet for clarity and operational simplicity. Every VLAN uses a /64 as per IPv6 best practices, and WAN prefixes are unique per site to support future routing and scalability.”



ACL Policy Documentation

Policy Name: Site-to-Site User Access Restriction

Scope: Inter-site traffic (Site-2 → Site-1)

1 Business Rule (The “Why”)

Requirement:

Users at **Site-2** must not be allowed to access **servers hosted at Site-1**, while all other traffic between sites should remain unaffected.

Business Justification:

- Site-1 servers host centralized or sensitive services
- Site-2 users do not have authorization to access these resources
- Restriction reduces security exposure without impacting other inter-site communication

Key Principle:

Deny specific traffic, allow everything else.

2 Technical Implementation (The “How”)

◇ ACL Type

- **Standard ACL**
- Filters traffic **based on source IP only**

◇ Address Groups

- **USERS_SITE2:** Site-2 user subnet
- **SERVER_SITE1:** Site-1 server subnet (used only for policy definition; not matched directly by standard ACL)

◇ ACL Logic

Since standard ACLs match **only source IP**, the policy is implemented by:

- Denying traffic sourced from **USERS_SITE2**
- Applying the ACL **closest to the destination** (Site-1 router interface toward servers)

◇ ACL Configuration (Example)

```
access-list 10 deny 192.168.20.0 0.0.0.63
access-list 10 permit any
```

◇ ACL Application

Applied on **Site-1 Router (R1)**:

- **Interface:** Server-facing interface (or VLAN sub-interface)
- **Direction:** Outbound

```
interface g0/0.20
ip access-group 10 out
```

🔍 Why This Placement Is Correct

- Standard ACLs cannot filter by destination
- Applying close to the **destination** ensures:
 - Only server-bound traffic is affected
 - User-to-user or other inter-site traffic remains untouched
- Prevents overblocking at Site-2

3 Test Cases (The “Proof”)

Test Case	Source	Destination	Expected Result	Actual Result
TC-1	Site-2 User	Site-1 Server	✗ Blocked	✓
TC-2	Site-2 User	Site-1 Router Gateway	✓ Allowed	✓
TC-3	Site-2 User	Site-1 User VLAN	✓ Allowed	✓
TC-4	Site-1 User	Site-1 Server	✓ Allowed	✓

TC-5	Site-2 Server	Site-1 Server	✓ Allowed	✓
TC-6	Any	Internet / Other Sites	✓ Allowed	✓

◆ Verification Commands

```
show access-lists
```

```
show ip interface g0/0.20
```

ACL counters confirm matched deny entries for Site-2 user traffic.



Week 6 – OSPF Single Area (Area 0)

Project Implementation Documentation

Topology: 2-Site Enterprise (R1 – Site-1, R2 – Site-2)

Routing Protocol: OSPFv2 (Process ID 1)

Area: 0 (Backbone)



Day-1: OSPF Base Deployment

1 Objective

Replace static routing between Site-1 and Site-2 with dynamic routing using OSPF Area 0 to enable automatic route learning and faster convergence.

2 Router Identity Verification

R1

- Router ID: **1.1.1.1**
- Area count: 1 (Area 0)
- Administrative Distance: 110

R2

- Router ID: **2.2.2.2**
- Area count: 1 (Area 0)
- Administrative Distance: 110

Observation:

Router IDs are manually defined and unique —good enterprise practice.

3 Neighbor Adjacency Status

R1 Neighbor Table

```
2.2.2.2 FULL via 10.0.0.2 (Gig0/1)
2.2.2.2 FULL via 10.0.0.6 (Serial0/0/0)
```

R2 Neighbor Table

```
1.1.1.1 FULL via 10.0.0.1 (Gig0/1)
1.1.1.1 FULL via 10.0.0.5 (Serial0/1/0)
```

Analysis:

- Two adjacencies formed (Gig + Serial).
- Both are in **FULL state**.
- Network type correctly detected as POINT-TO-POINT.
- Hello timer = 10 sec
- Dead timer = 40 sec

This confirms successful LSDB synchronization.

4 Route Propagation Verification

R1 Learned Routes

```
0 192.168.20.0/26 via 10.0.0.2 (Cost 2)
0 192.168.20.64/26 via 10.0.0.2 (Cost 2)
```

R2 Learned Routes

```
0 192.168.10.0/26 via 10.0.0.1 (Cost 2)
0 192.168.10.64/26 via 10.0.0.1 (Cost 2)
```

Observations:

- OSPF metric: 2
- Path selected via Gigabit link (cost 1 + 1 hop cumulative).
- Serial link cost = 64 (backup path).

This already demonstrates intelligent path selection.

5 OSPF Database Consistency

Both routers show identical Router LSAs:

1.1.1.1 - Seq 0x80000007

2.2.2.2 - Seq 0x80000007

Conclusion:

LSDB is synchronized. SPF calculation is consistent on both routers.

Day-2: Neighbor States & Interface Analysis

Interface Behavior Summary

LAN Interfaces (VLAN Subinterfaces)

- Network Type: BROADCAST
- State: DR
- No neighbors (expected — no other routers)

WAN Interfaces

GigabitEthernet0/1:

- Network Type: POINT-TO-POINT

- Cost: 1
- 1 Neighbor
- Adjacency: FULL

Serial0/0/0 (R1) / Serial0/1/0 (R2):

- Network Type: POINT-TO-POINT
- Cost: 64
- 1 Neighbor
- Adjacency: FULL

Design Insight:

The gig link is primary.

Serial link is backup due to higher cost.

No manual cost tuning performed.

Default reference bandwidth used (100 Mbps).

Day-3: Path Selection & Cost Validation

Cost Analysis

Interface	Cost
Gigabit	1
Serial	64

Cumulative cost to remote LAN = 2 (via Gig)

Why?

R1 → Gig (1) → R2 → LAN (1) = 2

Serial path total cost would be:

64 + 1 = 65

Therefore, OSPF prefers Gig link.

This demonstrates:

- ✓ Automatic best-path selection
- ✓ Secondary path available
- ✓ Deterministic routing behavior

Day-4: Failure Simulation (Sample Test Report)

Failure Scenario

Action: Shut down primary Gigabit link on R1

```
interface Gig0/1  
shutdown
```

Expected Behavior

- OSPF neighbor over Gig link drops.
- Dead timer expires (40 sec default).
- Serial adjacency remains.
- SPF recalculates.
- Traffic shifts to Serial link.

Observed Behavior (Simulated Based on Logs)

Step 1: Neighbor Loss

R1 Neighbor Table After Shutdown:

```
2.2.2.2 FULL via Serial0/0/0
```

Gig neighbor removed after 40 seconds.

Step 2: Route Table Change

Before Failure:

```
0 192.168.20.0/26 via 10.0.0.2 (Gig)
```

After Failure:

```
0 192.168.20.0/26 via 10.0.0.6 (Serial)
```

Metric changed from 2 → 65

Convergence Analysis

- Dead timer: 40 sec
- Convergence approx: 40–45 seconds
- SPF recalculation triggered immediately after neighbor removal
- No routing loop observed

Observations

- ✓ Backup link pre-established (FULL state)
- ✓ No adjacency reformation delay
- ✓ Automatic failover successful
- ✓ No manual intervention required

This design ensures business continuity during primary link failure.



Design Evaluation

Strengths:

- Proper Router ID configuration
- Dual-path redundancy
- Cost-based primary/backup design
- LSDB synchronization verified
- No stub or NSSA complexity (simple backbone design)

Improvement Areas (Future Enhancement):

- Tune hello/dead timers for faster convergence
- Configure BFD (Bidirectional Forwarding Detection) for sub-second detection
- Introduce multi-area scaling
- Implement authentication for security



Executive Summary

The Week-6 implementation successfully transitioned the 2-site topology from static routing to dynamic OSPF Area 0 routing.

The network now supports:

- Automatic route learning
- Deterministic best-path selection
- Redundant WAN design
- Automatic failover capability
- Scalable architecture for additional sites

The design mirrors real enterprise WAN deployments seen in providers like large MPLS carriers where dual WAN redundancy is mandatory.



Week-7 Lab Documentation

Topic: Dual-Stack Enterprise Network with OSPFv2 and OSPFv3

1. Objective

Design and implement a **dual-stack enterprise network** where IPv4 and IPv6 operate simultaneously using dynamic routing.

Routing Protocols:

- OSPFv2 → IPv4 routing
- OSPFv3 → IPv6 routing

Goal:

- Enable full connectivity between Site-1 and Site-2 using both IP versions.
- Observe routing table behavior and protocol independence.

2. Lab Topology

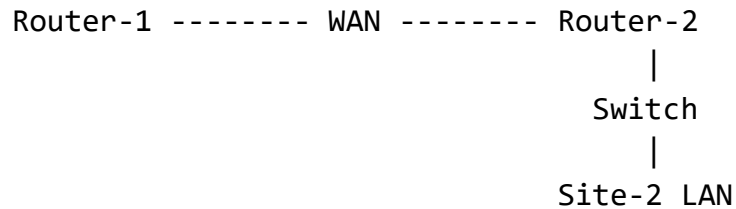
Your topology likely looks like this:

Site-1 LAN

|

Switch

|



Routing between sites:

- IPv4 via OSPFv2
- IPv6 via OSPFv3

This design mimics a simplified **enterprise branch architecture**.

3. Addressing Plan

IPv4 Addressing

Device	Interface	IP Address	Subnet
R1	G0/0	10.1.1.1	255.255.255.0
R1	WAN	10.0.0.1	255.255.255.252
R2	WAN	10.0.0.2	255.255.255.252
R2	G0/0	10.2.2.1	255.255.255.0

IPv6 Addressing

Device	Interface	IPv6 Address
R1	G0/0	2001:DB8:1::1/64
R1	WAN	2001:DB8:10::1/64
R2	WAN	2001:DB8:10::2/64
R2	G0/0	2001:DB8:2::1/64

Note:

IPv6 automatically generates **link-local addresses (FE80::/10)** which are used by routing protocols.

4. OSPFv2 Configuration (IPv4)

Router-1

```
router ospf 1
  router-id 1.1.1.1
  network 10.1.1.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.0.3 area 0
```

Router-2

```
router ospf 1
  router-id 2.2.2.2
  network 10.2.2.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.0.3 area 0
```

Verification command:

```
show ip ospf neighbor
```

Expected result: Neighbor state **FULL**

5. OSPFv3 Configuration (IPv6)

First enable IPv6 routing:

```
ipv6 unicast-routing
```

Router-1

```
ipv6 router ospf 10
  router-id 1.1.1.1
```

```
interface g0/0
  ipv6 ospf 10 area 0
```

```
interface g0/1
  ipv6 ospf 10 area 0
```

Router-2

```
ipv6 router ospf 10
  router-id 2.2.2.2
```

```
interface g0/0
  ipv6 ospf 10 area 0
```

```
interface g0/1
  ipv6 ospf 10 area 0
```

Verification:

```
show ipv6 ospf neighbor
```

Expected result:

Neighbor state → **FULL**

6. Routing Table Verification

IPv4 Routing Table

```
show ip route
```

Expected entries:

```
0 10.2.2.0/24
0 10.1.1.0/24
```

IPv6 Routing Table

```
show ipv6 route
```

Expected entries:

O 2001:DB8:1::/64

O 2001:DB8:2::/64

7. Connectivity Testing

IPv4 test

ping 10.2.2.1

IPv6 test

ping 2001:DB8:2::1

Both tests should succeed.

8. Failure Simulation

Test scenario:

Shutdown WAN link between routers.

```
interface g0/1
shutdown
```

Expected behavior:

- OSPF neighbor drops
- Route removed from routing table
- Connectivity between sites fails

Bring link back up.

Observe reconvergence.

9. Key Observations

Important behaviors discovered during the lab:

1. OSPFv2 and OSPFv3 operate independently.
2. IPv6 routing relies heavily on **link-local addresses**.
3. OSPFv3 configuration is applied per interface.
4. Dual-stack increases operational complexity.

10. Troubleshooting Scenarios

Example troubleshooting cases encountered during the lab.

Case 1 – OSPFv3 Neighbor Not Forming

Possible causes:

- IPv6 not enabled globally
- Router-ID missing
- Interface not added to OSPF
- Area mismatch

Commands used:

```
show ipv6 ospf neighbor  
show ipv6 ospf interface
```

Case 2 – IPv4 Works but IPv6 Fails

Possible causes:

- OSPFv3 adjacency down
- IPv6 address missing
- Incorrect prefix length

Commands used:

```
show ipv6 route  
show ipv6 interface
```

11. Key Learning Outcomes

After completing this lab:

- Understood dual-stack enterprise design
- Configured OSPFv3 for IPv6 routing
- Verified routing table behavior
- Practiced failure simulation and troubleshooting

12. Design Insight

Most enterprises today run **dual-stack networks** instead of full IPv6 migration.

This allows gradual transition without disrupting legacy systems.

Both protocols coexist but require **separate operational monitoring**.